

Orbits Close to Asteroid 4769 Castalia

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We use a radar-derived physical model of 4769 Castalia (1989 PB) to investigate close orbit dynamics around that kilometer-sized, uniformly rotating asteroid. Our methods of analysis provide a basis for systematic studies of particle dynamics close to any uniformly rotating asteroid.

We establish that a Jacobi integral exists for particles orbiting this asteroid, examine the attendant zero-velocity surfaces, find families of periodic orbits, and determine their stability. All synchronous orbits and direct orbits within ~ 3 mean radii of Castalia are unstable and are subject to impact or escape from Castalia. Retrograde orbits are mostly stable and allow particles to orbit close to the asteroid surface.

We derive a model which allows us to predict the escape conditions of a particle in orbit about Castalia and the (temporary) capture conditions for a hyperbolic interloper. Orbits within 1.5 km of Castalia are subject to immediate ejection from the system. Hyperbolic orbits with a $V_{\infty} < 0.4$ m/sec can potentially be captured by Castalia if their periapsis radius is within ~ 2 km. For Castalia this capture region is small, but the results also apply to larger asteroids whose capture regions would also be larger.

We determine bounds on ejecta speeds which either ensure ejecta escape or re-impact as functions of location on Castalia's surface. The speeds that ensure escape range from 0.28 to 0.84 m/sec and the speeds that ensure re-impact range from 0 to 0.18 m/sec. Speeds between these two bounds lead either to escape, re-impact, or potentially finite-time stable orbits. We develop a simple criterion which can establish whether a particle could have been ejected from the asteroid in the past and if it will impact the surface in the future.

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1. INTRODUCTION

A detailed understanding of the dynamics of orbits close to small (kilometer-sized), irregularly shaped asteroids is

required for realistic investigation of the systematics of cratering ejecta and the evolution of those objects' regoliths. The stability of close orbits bears on questions about possible satellites of small bodies and on the practical challenge of maneuvering spacecraft near the surfaces of small bodies. Several authors have touched on the issue of close-orbit dynamics (Chauvineau *et al.* 1993, Cintala *et al.* 1979, Davis *et al.* 1981, Dobrovolskis and Burns 1980, Scheeres 1994, 1995b, Weidenshilling *et al.* 1989, Harris 1987, Geisler *et al.* 1995, Petit *et al.* 1994). However, the previous literature has used simplistic shape models in general (ellipsoids) or has not dealt with the full nature of the dynamics of particles near an irregular shape.

The intent of this paper is to provide a detailed look at the orbital dynamics associated with asteroid 4769 Castalia (1989 PB), whose shape has been determined by radar (Hudson and Ostro, 1994), and to develop general analysis tools which can describe these dynamics in a meaningful way. Section 2 describes the generation and description of the Castalia shape, rotational dynamics, and density model. Then we state the general equations of motion of a particle about such an asteroid, discuss the integrals of motion of these equations, and express the dynamic equations in terms of the osculating elements.

Section 3 derives dynamical results for the shape model. First we calculate zero-velocity surfaces of the asteroid and explain the relevant information which can be obtained from them. Next we compute families of periodic orbits and determine their stability, obtaining general insights into particle motion in various regimes about the asteroid. Then we derive some general results on the dynamics of near-equatorial orbits. This analysis enables us to estimate under what conditions a passing hyperbolic particle could become (temporarily) captured by Castalia, and conversely

under what conditions an orbiting particle could be ejected from the Castalia system. For completeness we review the dynamics of particles in retrograde orbits. It is seen that such orbits may be qualitatively described using simple formulas.

Section 4 deals with issues associated with trajectories of cratering ejecta and regolith evolution in general. First we discuss the local slopes on the Castalia surface in the context of the gravity field and centripetal accelerations. Then we develop a criterion which can discriminate whether or not an orbiting particle could have originated from the surface of Castalia. This criterion is useful in establishing whether an orbiting particle could be ejecta or if it might impact the asteroid surface in the future. We also characterize the evolution of ejecta from the surface of Castalia in this section. To this end, estimates of the speed above which ejecta will definitely escape the Castalia system as well as estimates of the speed below which ejecta will definitely fall back onto the surface of Castalia are defined as a function of position on Castalia's surface. Finally we examine the possibility of ejecta being trapped in stable orbits and draw conclusions about the likely distribution of return ejecta on the surface.

Some of our specific orbital results have been recorded in video format (de Jong and Suzuki 1996). This video depicts several different particle orbits in inertial, Castalia-fixed, and particle-fixed viewpoints. It is useful for visualizing the dynamics of particles about a small asteroid such as Castalia. Included in the video are segments depicting ejecta return and escape orbits, stable retrograde orbits, unstable polar periodic orbits (at varying densities), and a hypothetical landing orbit for a powered spacecraft.

2. CASTALIA MODEL AND GRAVITY FIELD

Hudson and Ostro (1994) used techniques first described by Hudson (1993) to invert delay-Doppler radar images of Castalia that had been obtained at Arecibo in 1989 by Ostro *et al.* (1990). Their inversion yielded a 167-parameter, 3-D shape model that is bifurcated into two distinct, irregular, kilometer-sized lobes separated by a crevice that has an average depth of at least 100 m and is oriented nearly perpendicular to the asteroid's dimension. This inversion assumed uniform density and principal-axis rotation to ensure plausible extrapolation of the reconstructed surface into the (polar) region not seen by the radar. In this paper we adopt the shape model of Hudson and Ostro as a global estimate of the shape of Castalia realizing that quantitative results stated for the southern regions are not well-constrained but are merely plausible estimates. The asteroid's rotation period, estimated from the radar images and also from optical lightcurves, is 4.07 hr. The model's volume is 0.671 km^3 , which corresponds to a mean radius of 0.543

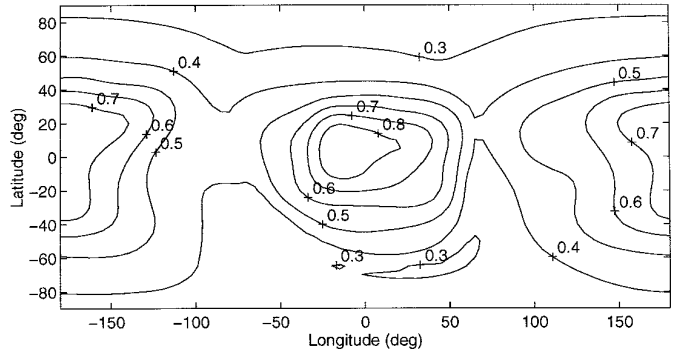


FIG. 1. Radius contours of Castalia.

km. A contour plot of the radius of Castalia is shown in Fig. 1.

For describing positions about Castalia we use a body-fixed coordinate system whose origin is at the model's centroid and whose axes (x , y , z) correspond to the principal axes of smallest, intermediate, and largest moment of inertia, respectively. The model's rotation pole lies along the z axis and its shape fits into the bounding box, centered at its center of mass, $-0.762 \leq x \leq 0.851$, $-0.462 \leq y \leq 0.519$, $-0.444 \leq z \leq 0.382$. The ratios of the moments of inertia are

$$I_x/I_z = 0.37207 \quad (1)$$

$$I_y/I_z = 0.93805. \quad (2)$$

An estimate of Castalia's density is derived using the following argument. First, estimates of Castalia's radar cross section and polarization ratio, when interpreted in the context of the Hudson and Ostro (1994) modeling (see footnote 13 of that reference), suggest that the asteroid's radar albedo due to single back reflections from smooth surface elements is within 40% of 0.12. That albedo is a first approximation to the Fresnel power-reflection coefficient R , which in this situation can be related to surface bulk density ρ through empirical formulas like (Ostro *et al.* 1985)

$$\rho(R) = (R + 0.13)/0.12. \quad (3)$$

This logic leads to an interval, $1.7 \leq \rho \leq 2.5 \text{ g/cm}^3$, for the smooth component of Castalia's surface. We adopt a nominal value of 2.1, which corresponds to respective porosities of 60% and 40% for ordinary chondrites and stony irons, the candidate meteorite analogs for S class asteroids like Castalia.

That density gives a total mass

$$M = 1.4091 \times 10^{12} \text{ kg} \quad (4)$$

and a gravitational parameter

$$\mu = GM = 9.40 \times 10^{-8} \text{ km}^3/\text{sec}^2, \quad (5)$$

where $G = 6.67259 \times 10^{-20} \text{ km}^3/\text{kg}/\text{sec}^2$.

There are two practical approaches for generating the corresponding constant density gravitational field. The more common one determines the coefficients of a harmonic expansion of the gravity field. As is known classically (MacMillan 1930), the gravitational field of any arbitrary body may be expressed as an infinite series expansion. The determination of these series coefficients is performed by repeatedly integrating over the entire volume of the shape, one integration for each coefficient. If constant density is assumed, these volume integrations may be reduced to integrations over the surface of the body. The most important terms of the harmonic expansion of the gravity field correspond to the C_{20} and C_{22} coefficients. For the Castalia model these are found to be

$$C_{20} = -7.275 \times 10^{-2}/r_o^2 \quad (6)$$

$$C_{22} = 2.984 \times 10^{-2}/r_o^2, \quad (7)$$

where r_o is an arbitrary normalization radius. In the Appendix is a brief description of the gravity field expansion and a table giving the gravitational coefficients up to order 4. The computations used in this paper use gravitational coefficients up to order 16. The approximation error is usually small if sufficient terms are kept. If the gravity field is evaluated close to the maximum radius of the body (0.85 km) then the error will become larger as the higher order gravitational terms will then carry relatively more weight. When evaluated within the maximum radius the harmonic expansion no longer converges globally. In some instances the accuracy may still be sufficient, but usually when the series is evaluated close to the surface of the body the divergence in the series is too severe to yield acceptable answers.

Another approach is to use a polyhedral shape approximation and the exact closed form expression of the gravitational field of a polyhedron (Werner 1994). This approach is especially attractive when evaluation of the gravity field must be made close to or on the surface of the asteroid. This formulation satisfies Laplace's equation exterior to the body and Poisson's equation in the interior of the body and thus is globally valid. It is also the exact potential for the given shape (assuming constant density). The disadvantage of this approach is the computational time required. Every time the potential (or accelerations) are evaluated, a summation over every face of the polyhedral shape must be performed. This is equivalent to computing one coefficient in the harmonic expansion. Thus, if the accelerations are to be computed over some time span, it is clearly more efficient to use a harmonic expansion. However, this formulation is well suited to running on a parallel computer.

3. EQUATIONS OF MOTION AND CONSERVED QUANTITIES

Given a gravitational field and the rotational state of the model, the equations of motion may be written. The most efficient expression of these equations is given in the body-fixed frame of the asteroid. Given an integrated result in this frame, it is easy to transform to the inertial frame if desired. The benefit of this formulation is that no rotational transformations must be made to evaluate the gravitational accelerations and that the equations of motion are time invariant for a uniformly rotating central body.

For the general rotation of an asteroid, the body fixed equations of motion for a small particle are (Greenwood 1988)

$$\ddot{\mathbf{r}} + 2\boldsymbol{\Omega} \times \dot{\mathbf{r}} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) + \dot{\boldsymbol{\Omega}} \times \mathbf{r} = U_{\mathbf{r}}, \quad (8)$$

where $\mathbf{r}$ is the body-fixed vector from the asteroid center of mass, $(\dot{})$ and $(\ddot{})$ are first and second time derivatives with respect to the body-fixed, rotating frame, $\boldsymbol{\Omega}$ is the instantaneous rotation vector of the asteroid with magnitude $|\boldsymbol{\Omega}| = \omega$, and $U_{\mathbf{r}}$ is the gradient of the gravitational potential $U(\mathbf{r})$ which is time-invariant in a body-fixed frame.

To analyze the dynamical equations it is of interest to find conserved quantities, or integrals. Define the function J as

$$J = \frac{1}{2} \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} - \frac{1}{2} (\boldsymbol{\Omega} \times \mathbf{r}) \cdot (\boldsymbol{\Omega} \times \mathbf{r}) - U(\mathbf{r}), \quad (9)$$

where all the quantities are as defined previously. Take the time derivative of this function, performing all differentiations with respect to the body-fixed frame

$$\dot{J} = \dot{\mathbf{r}} \cdot \ddot{\mathbf{r}} - (\boldsymbol{\Omega} \times \mathbf{r}) \cdot (\dot{\boldsymbol{\Omega}} \times \mathbf{r}) - (\boldsymbol{\Omega} \times \mathbf{r}) \cdot (\boldsymbol{\Omega} \times \dot{\mathbf{r}}) - U_{\mathbf{r}}(\mathbf{r}) \cdot \dot{\mathbf{r}}. \quad (10)$$

Next, take the dot product of Eq. (8) with $\dot{\mathbf{r}}$ and re-arrange the results:

$$\dot{\mathbf{r}} \cdot \ddot{\mathbf{r}} - (\boldsymbol{\Omega} \times \mathbf{r}) \cdot (\boldsymbol{\Omega} \times \dot{\mathbf{r}}) + \dot{\mathbf{r}} \cdot (\dot{\boldsymbol{\Omega}} \times \mathbf{r}) - U_{\mathbf{r}}(\mathbf{r}) \cdot \dot{\mathbf{r}} = 0. \quad (11)$$

Combine Eqs. (10) and (11) to obtain

$$\dot{J} = \dot{\boldsymbol{\Omega}} \cdot (\mathbf{v}_I \times \mathbf{r}) \quad (12)$$

$$\mathbf{v}_I = \dot{\mathbf{r}} + \boldsymbol{\Omega} \times \mathbf{r}, \quad (13)$$

where $\mathbf{v}_I$ is the inertial velocity of the test particle. If the asteroid is in principal axis rotation (i.e., $\dot{\boldsymbol{\Omega}} \equiv 0$) then the quantity J is conserved, and is in fact the Jacobi integral

for the equations of motion. Thus uniform rotation of the asteroid implies that an integral of the motion exists. An equivalent statement is that the equations of motion (in the body-fixed frame) are time invariant when $\dot{\Omega} \equiv 0$.

Since Castalia is a uniformly rotating asteroid, henceforth we assume the Jacobi function to be constant unless otherwise noted. Thus, given an initial particle position and velocity, the Jacobi function $J(\mathbf{r}, \dot{\mathbf{r}})$ is constant for all ensuing motion of that particle. There are a variety of initial conditions which may lead to the same constant value of the Jacobi function, so it is useful to define the Jacobi constant C such that

$$J(\mathbf{r}, \dot{\mathbf{r}}) + C = 0. \quad (14)$$

For the Castalia model (or any uniformly rotating asteroid) the equations of motion in scalar form reduce to

$$\ddot{x} - 2\omega\dot{y} = \omega^2 x + U_x \quad (15)$$

$$\ddot{y} + 2\omega\dot{x} = \omega^2 y + U_y \quad (16)$$

$$\ddot{z} = U_z. \quad (17)$$

The Jacobi constant C is explicitly calculated as

$$C = V(x, y, z) - T_E, \quad (18)$$

where

$$V(x, y, z) = \frac{1}{2} \omega^2 (x^2 + y^2) + U(x, y, z) \quad (19)$$

is the modified potential and

$$T_E = \frac{1}{2} (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) \quad (20)$$

is the kinetic energy of the particle with respect to the rotating asteroid.

Next recall the standard definition of the Keplerian (or two-body) energy

$$C_2 = \frac{1}{2} \mathbf{v}_I \cdot \mathbf{v}_I - \frac{\mu}{|\mathbf{r}|}, \quad (21)$$

where μ is the gravitational parameter of the attracting body and $|\mathbf{r}|$ is the Euclidian norm of the particle position vector. Let us express this quantity in the body-fixed, rotating reference frame using the substitution $\mathbf{v}_I = \dot{\mathbf{r}} + \Omega \times \mathbf{r}$. Also, assume that the gravity force potential U of the asteroid is expressed as

$$U(\mathbf{r}) = \frac{\mu}{|\mathbf{r}|} + \sum_i U_i(\mathbf{r}), \quad (22)$$

where the U_i terms correspond to all the higher degree and order harmonics of the general gravity field. Then the Keplerian energy is

$$C_2 = \frac{1}{2} (\dot{\mathbf{r}} + \Omega \times \mathbf{r}) \cdot (\dot{\mathbf{r}} + \Omega \times \mathbf{r}) - U(\mathbf{r}) + \sum_i U_i(\mathbf{r}). \quad (23)$$

Substitution of the formal Jacobian integral (Eq. 14) into the above equation yields

$$C_2 = \sum_i U_i(\mathbf{r}) + (\Omega \times \mathbf{r}) \cdot (\dot{\mathbf{r}} + \Omega \times \mathbf{r}) - C, \quad (24)$$

where $C = -J$ is the Jacobian constant. Take the time differential of the Keplerian energy to find

$$\dot{C}_2 = \sum_i \nabla U_i(\mathbf{r}) \cdot \dot{\mathbf{r}} + [\dot{\mathbf{r}} + 2\Omega \times \mathbf{r}] \cdot (\Omega \times \mathbf{r}). \quad (25)$$

Substitute in Eqs. (8), noting that $\nabla (\mu/|\mathbf{r}|) = -\mu\mathbf{r}/|\mathbf{r}|^3$, to find the final result

$$\dot{C}_2 = \sum_i \nabla U_i(\mathbf{r}) \cdot \mathbf{v}_I \quad (26)$$

$$\mathbf{v}_I = \dot{\mathbf{r}} + \Omega \times \mathbf{r}, \quad (27)$$

where the inertial velocity vector $\mathbf{v}_I$ may also be expressed as

$$\mathbf{v}_I = \frac{er^2 \dot{f} \sin f}{a(1-e^2)} \hat{\mathbf{r}} + r \dot{f} \hat{\mathbf{v}}_T,$$

with $\hat{\mathbf{v}}_T$ the unit component of the velocity normal to the radius vector. This result may also be restated in the more intuitive form

$$\dot{C}_2 = \sum_i U'_i(\mathbf{r}), \quad (29)$$

where the U'_i term denotes the time derivative of U_i with respect to the inertial frame.

In terms of the osculating Keplerian elements, the two-body energy is

$$C_2 = -\frac{\mu}{2a}, \quad (30)$$

where a is the osculating semi-major axis of the orbit.

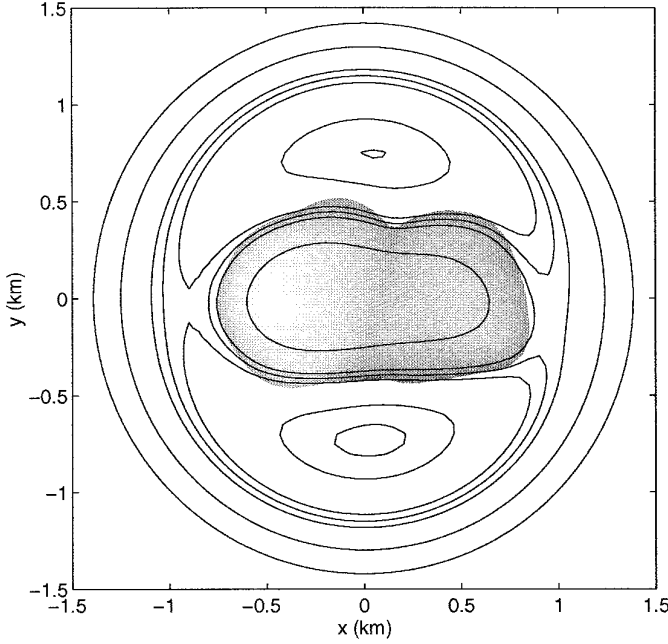


FIG. 2. Zero-velocity curve in the $x, y, z = 0$ plane. Visible are 4 regions where the curves intersect or close in on themselves; these regions contain isolated equilibrium points.

Taking the time derivative of this and using Eq. (29) results in

$$\dot{a} = \frac{2}{n^2 a} \sum_i U'_i, \quad (31)$$

where n is the mean motion of the orbit ($n = \sqrt{\mu/a^3}$). One also has from the Lagrange planetary equations the result (Kaula 1966)

$$\dot{a} = \frac{2}{na} \frac{\partial \sum_i U_i}{\partial M},$$

where M is the mean anomaly of the orbit. From Eqs. (31) and (32) we find

$$\frac{\partial \sum_i U_i}{\partial M} = \frac{1}{n} \sum_i U'_i(\mathbf{r}) \quad (33)$$

which is useful as it is sometimes desired to avoid the mean anomaly in deriving explicit forms for the Lagrange equations.

Equations for the osculating periaapsis and apoapsis can be derived to be (Kaula 1966)

$$\dot{r}_p = -\frac{(1-e)^2}{nae} \sum_i U_{iM} + \frac{\sqrt{1-e^2}}{nae} \sum_i U_{iv} \quad (34)$$

$$\dot{r}_a = \frac{(1+e)^2}{nae} \sum_i U_{iM} - \frac{\sqrt{1-e^2}}{nae} \sum_i U_{iv}, \quad (35)$$

where ν is the argument of periapsis.

4. Dynamics about Castalia

In the absence of solar perturbations, the orbital dynamics of a particle far from Castalia approach those for the classical case of motion about an oblate planet (Scheeres 1994). Close to Castalia, however, this comparison is no longer valid and the orbits may exhibit unstable and even chaotic behavior (Chauvineau *et al.* 1993, Scheeres 1994). There are several approaches that can illuminate the dynamics and the general structure of phase space in this situation.

4.1. Zero-Velocity Surfaces

Zero-velocity surfaces, defined using the Jacobi integral, provide concrete information regarding the possible motion of a particle. Since $T_E \geq 0$ it is possible to define an inequality

$$V(x, y, z) \geq C \quad (36)$$

that partitions the x, y, z space into regions where the

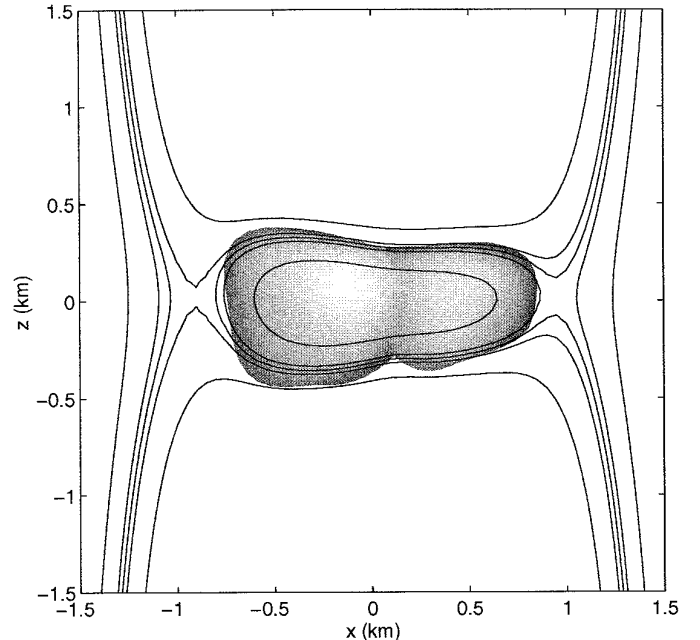


FIG. 3. Zero-velocity curve in the $x, z, y = 0$ plane. Two equilibrium points exist at the long ends of the asteroid where the curves intersect.

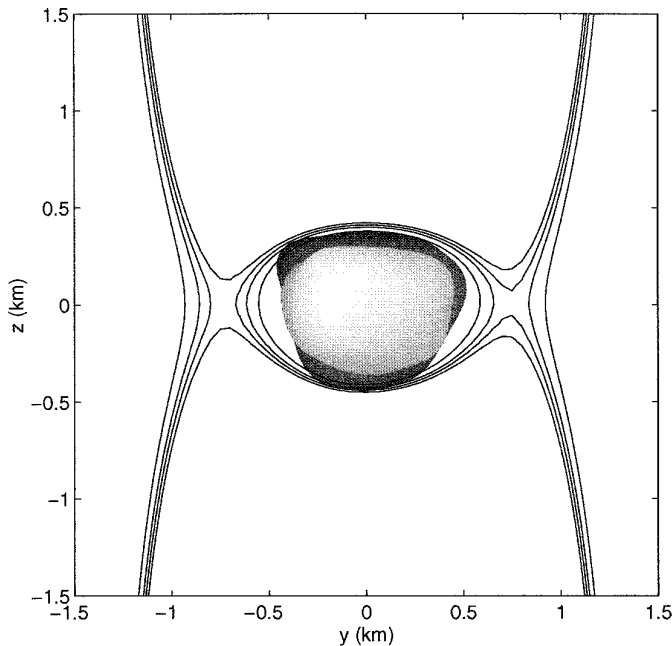


FIG. 4. Zero-velocity curve in the $y, z, x = 0$ plane. Two equilibrium points exist along the short ends of the asteroid where the curves intersect.

particle may be found and where it may not be found, given a specific value of C . Note that $V(x, y, z) \geq 0$ over the entire space. Thus, if $C < 0$, the inequality is identically satisfied and there are no *a priori* constraints on where the particle may be found.

If $C > 0$, there will be regions of space where the inequality is violated, and hence where no particle may travel. When these forbidden regions separate space into disjoint regions a particle can never travel between these regions. The general situation is discussed more fully in Scheeres (1994), where the central body is assumed to be a tri-axial ellipsoid.

Zero-velocity surfaces are defined by

$$V(x, y, z) = C. \quad (37)$$

This equation defines a 2-dimensional surface in the 3-dimensional x - y - z space. As the value of the Jacobi constant C is varied, the surfaces change. At critical values of C the surfaces intersect or close in upon themselves at points in the x - y - z space usually called equilibrium, or critical, points. Figures 2–4 are projections of the zero-velocity surface onto the $z = 0$, $y = 0$, and $x = 0$ planes, respectively. The surfaces are all evaluated close to the critical values of C . The zero-velocity curves presented here may be interpreted in much the same way as they

are interpreted for the restricted three-body problem (Hamilton and Burns 1991).

4.2. Synchronous Orbits and Stability

Clearly seen in Fig. 2 are four critical points, all of which lie near the equator and are separated by approximately 90° in longitude. At these points there is a net zero acceleration acting on the particle in the rotating frame, thus a particle placed there without initial velocity in the rotating frame will ideally remain in such an orbit indefinitely. These are truly circular orbits which are exactly synchronous with Castalia's rotation rate.

A more direct manner of computing these equilibrium points is to find all the solutions of $V_r(\mathbf{r}) = 0$. For a general gravity field there is no *a priori* number of solutions to this equation; the number of solutions depends on the shape and spin rate of the body. The number of solutions can be most easily found by observing the zero-velocity curves about the body in question and identifying all points where these curves intersect or bifurcate. In the case of Castalia we observe only four such solutions (Figs. 2–4). Call the synchronous orbits at the long ends of Castalia the $\pm S$ (ad-*dle*) solutions, and those at the shorter ends of Castalia the $\pm C$ (en-*ter*) solutions. For the given parameters of Castalia, the locations and Jacobi constant values of these orbits are

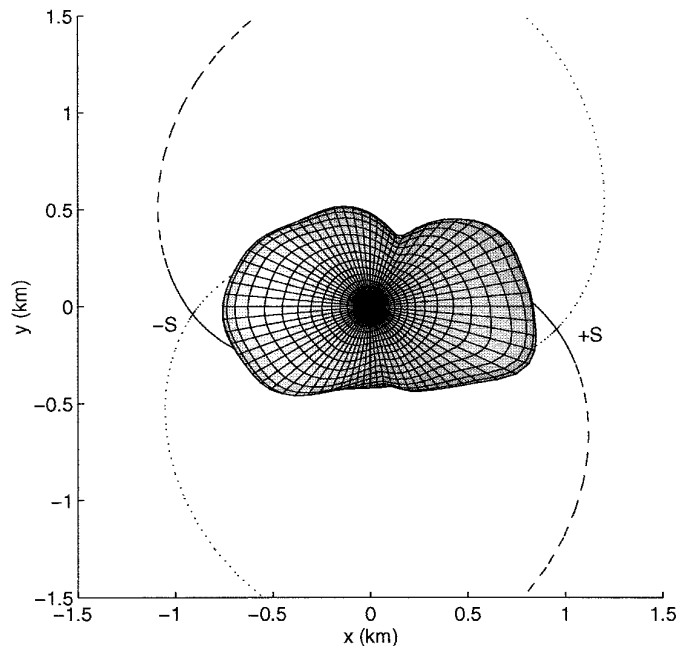


FIG. 5. Segments of the stable and unstable manifolds of the $\pm S$ equilibrium points. Stable manifolds are dotted lines, and unstable manifolds are dashed lines. These manifolds either escape from the asteroid or intersect the asteroid.

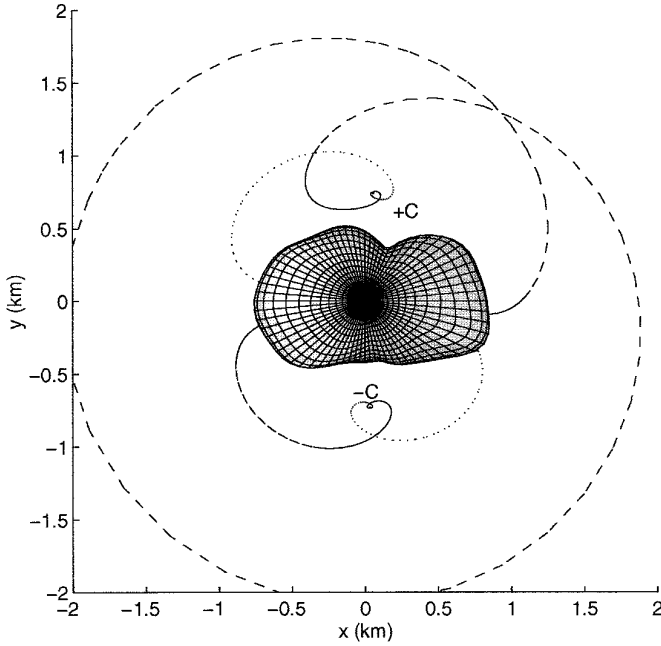


FIG. 6. Particular trajectories from the stable and unstable manifolds of the $\pm C$ equilibrium points. Stable manifolds are dotted lines, and unstable manifolds are dashed lines. The evolution of these manifolds is complicated, although it is believed that the major portion of these manifolds intersect the asteroid.

$$\mathbf{r}_{+S} = (0.955, -0.118, 0.021) \text{ km};$$

$$C_{+S} = 2.022 \times 10^{-7} \text{ km}^2/\text{sec}^2$$

$$\mathbf{r}_{-S} = (-0.910, -0.039, 0.024) \text{ km};$$

$$C_{-S} = 1.954 \times 10^{-7} \text{ km}^2/\text{sec}^2 \quad (38)$$

$$\mathbf{r}_{+C} = (0.060, 0.739, 0.009) \text{ km};$$

$$C_{+C} = 1.678 \times 10^{-7} \text{ km}^2/\text{sec}^2$$

$$\mathbf{r}_{-C} = (0.035, -0.730, 0.005) \text{ km};$$

$$C_{-C} = 1.664 \times 10^{-7} \text{ km}^2/\text{sec}^2.$$

For Castalia all four synchronous orbits are unstable (this means that a particle perturbed slightly from any of these orbits will depart the equilibrium point exponentially). The $\pm S$ orbits are hyperbolically unstable. Thus, any particle displaced from these body-fixed points will depart from that point on a local hyperbola. Figure 5 shows the stable and unstable manifolds of these points. Note that each of these points has stable and unstable manifolds trapped near the asteroid as well as stable and unstable manifolds trapped away from the asteroid. Each manifold is a 1-dimensional object in phase space (which is a generic property of such hyperbolic unstable points).

The properties of these manifolds provide sharp conditions for the limiting case of a particle escape trajectory.

Suppose the particle is started from Castalia's surface with a Jacobi constant incrementally less than C_S and along a stable manifold. Then the particle will follow the stable manifold as it approaches the S equilibrium point, taking an arbitrarily long time to arrive at this point. Once in the neighborhood of the point, the particle will come under the influence of an unstable manifold. If it comes under the influence of the unstable manifold trapped close to the asteroid it will fall back onto the asteroid. If it comes under the influence of the unstable manifold trapped away from the asteroid, it will move away from the asteroid along this manifold and will never return to the asteroid surface. For a particle started with a Jacobi constant greater than or equal to C_S , this escape path will no longer exist, and all such particles will eventually fall back onto the asteroid surface.

The $\pm C$ orbits are complex unstable. Thus, any particle displaced from these orbits will depart from that point on a (locally) hyperbolic spiral. Each of these manifolds is a 2-dimensional object in phase space and cannot be traced out as the hyperbolic manifolds of the $\pm S$ orbits could be. Figure 6 shows particular trajectories from the stable and unstable manifolds of the $\pm C$ orbits. These manifolds tend to lie near the equatorial plane of Castalia, and cause the radius of the trajectory to oscillate in an unstable manner, thus causing either close approaches to, or impact with, the asteroid.

The instability of the $\pm C$ equilibrium points highlights

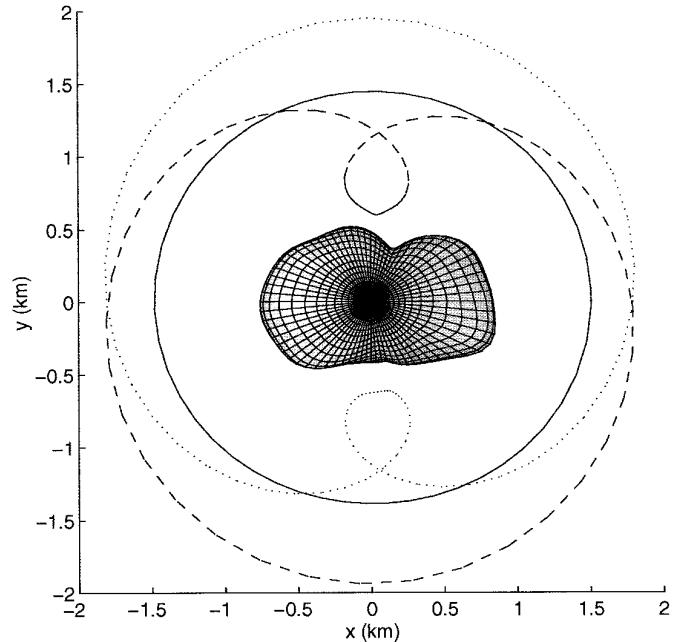


FIG. 7. Direct, quasi-equatorial periodic orbits about Castalia. Shown are three representative members. The solid line is a near-circular orbit, and the dashed and dotted lines are elliptic orbits with argument of periapsis 180° out of phase.

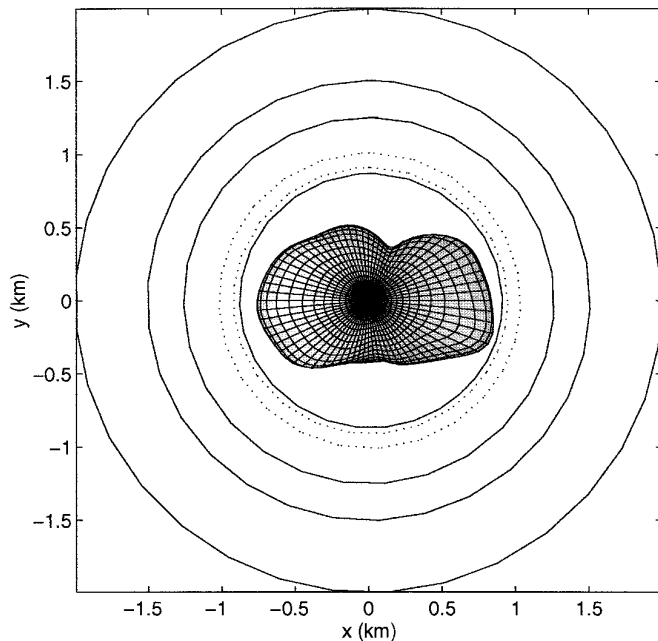


FIG. 8. Retrograde, quasi-equatorial periodic orbits about Castalia. All orbits of this family are nearly circular. The two dashed orbits bracket the interval of unstable orbits found for this family.

a classification that can be applied to Castalia (and to any uniformly rotating asteroid). The classification, as originally defined (Scheeres 1994), applies to any body with four synchronous orbits. If two of these orbits are unstable (the *S* orbits) and two of these orbits are stable (the *C* orbits), then the asteroid is classified as Type I. If all four of the orbits are unstable, as with Castalia, then the asteroid is classified as Type II. Note that the *S* orbits are always unstable.

4.3. Periodic Orbit Families

We also computed families of periodic orbits about the asteroid. The computation of these orbits requires a precision integration routine and a set of software tools which allows one to force the end-points of an orbit to coincide. The details of performing such a computation involves the use of Poincaré maps and Newton–Raphson iteration (Marchal 1990). Given one periodic orbit, other members in its family may be found via analytic continuation of the orbit with respect to some parameter, usually the Jacobi constant or the orbit period. For a nonsymmetric body such as Castalia, a coordinate plane is usually chosen to act

as the Poincaré surface. Some details of the computational procedure used here are given in (Scheeres 1995a).

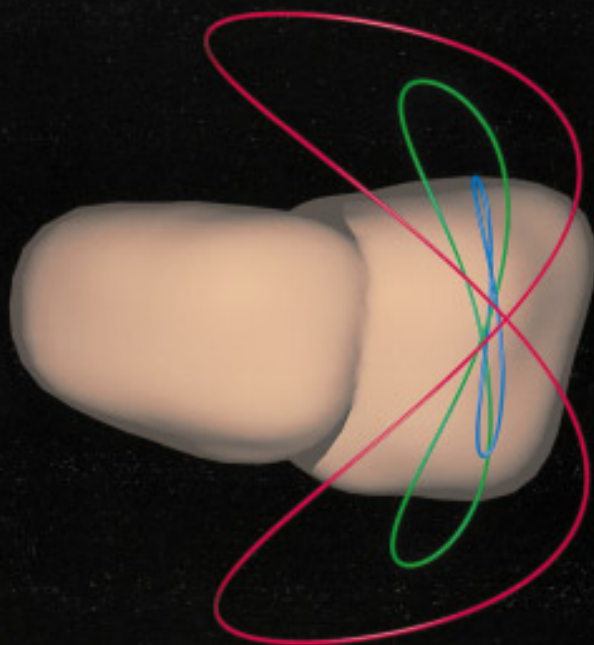
There are three main families of periodic orbits close to an asteroid such as Castalia: quasi-equatorial direct, quasi-equatorial retrograde, and nonequatorial. The retrograde orbits exist as quasi-circular orbits far from the asteroid and continue for decreasing radius until they intersect the asteroid itself. The direct orbits exist as quasi-circular orbits far from the asteroid and continue for decreasing radius until they approach the radii of the synchronous orbits. Then they split into elliptic orbits (with additional complications not studied here) and can be continued until they intersect the asteroid. The nonequatorial orbits exist only in the vicinity of certain radii where the out-of-plane (nodal) period of the orbit is commensurate with the rotation rate of the asteroid. In the case of 1:1 commensurability these orbits are continued from halo orbits associated with the four equilibrium points discussed above. As the orbital period of this family is varied, the inclination varies, approaching $i = 90^\circ$. The families associated with the 1:1 commensurability are all unstable. Other families may exist at different values of commensurability. These three families of orbits also exist in the case of a triaxial ellipsoid central body, except in that case they contain space symmetries. Note that these are, by no means, the only periodic orbit families about Castalia (or about a uniformly rotating asteroid in general). They are just the simplest families in terms of their geometry. This simplicity allows them to be used to characterize other, more general, orbits which are “close” to these families in phase space.

Figures 7–9 present members of each of these periodic orbit families in the body-fixed coordinate frame. Figure 7 presents some select members of the direct, quasi-equatorial family of periodic orbits about Castalia. This family begins as a family of quasi-circular orbits when far from Castalia (solid line) and then bifurcates into two families of elliptic orbits when close to Castalia (dashed and dotted lines). The two bifurcated branches of the family are close to being symmetric about the *y* axis, and would be if Castalia were symmetric about this axis.

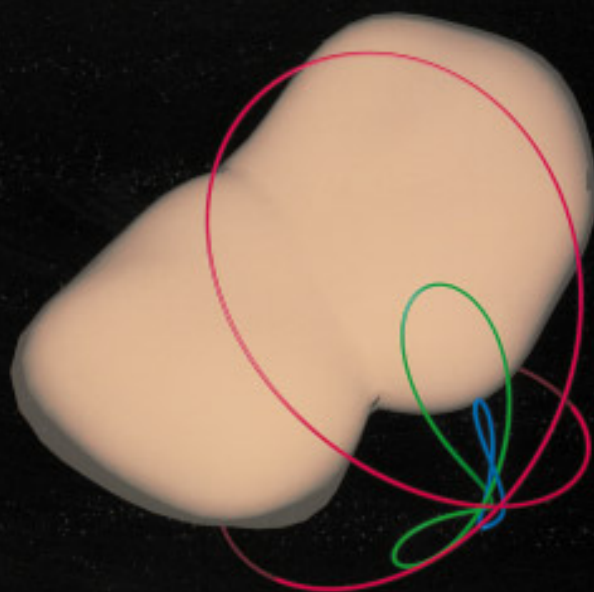
Figure 8 presents members of a family of retrograde, quasi-equatorial periodic orbits that also begins as a family of quasi-circular orbits far from Castalia, and remains essentially circular as it draws closer to the surface of Castalia. This family is stable up to grazing orbits, except for a small interval bounded by the two dashed lines in Fig. 8. The existence of the stable family indicates that it is possible for particles to orbit very close to the asteroid

FIG. 9. Three-dimensional views of nonequatorial periodic orbits. Shown are three members of this periodic orbit family. Note how the family progresses from a halo orbit about the equilibrium point to a near-polar orbit.

a



b



9

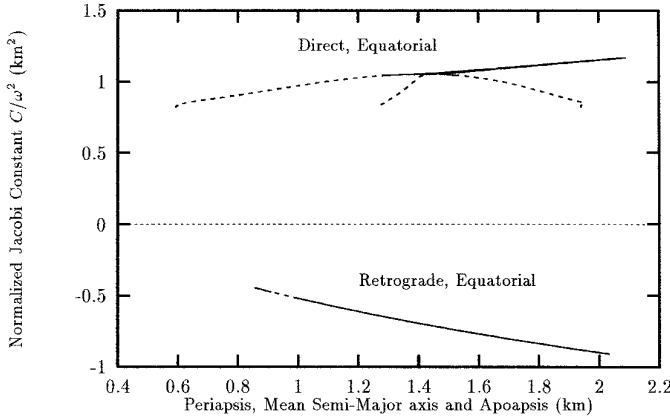


FIG. 10. Periodic orbit family stability. Plotted is the periapsis, mean semi-major axis, and apoapsis of the direct and retrograde quasi-equatorial orbit families. The solid lines indicate stable members, and the dashed lines indicate unstable members.

surface for extended periods of time (until third-body perturbations or tidal effects may perturb the orbit sufficiently).

Figure 9 depicts members of a periodic orbit family that is comprised of orbits with a non-zero inclination. Ideally, this family originates at the halo orbits associated with the $\pm C$ equilibrium points and terminates at the halo orbits associated with the $\mp S$ equilibrium points (or vice versa). For Castalia this family does not complete such a path, because it intersects with the longer ends of the asteroid as it moves from the $\pm C$ point to the $\mp S$ point. The figures give a 3-dimensional perspective of some select members of this family. A similar family also exists, ideally traversing from the $-C$ equilibrium point through a polar orbit to the $+S$ equilibrium point. For densities larger than 3 g/cm^3 this family will not intersect the asteroid. Due to the unstable nature of these orbits their manifolds permeate the phase space close to the asteroid surface, thus low speed ejecta from Castalia evolve under the influence of these manifolds. These manifolds will intersect with Castalia's surface or, barring that, will escape from the vicinity of the asteroid.

The stability of these periodic orbit families affects the nature of motion in the surrounding phase space. If a family is stable, then neighboring trajectories will oscillate about it and will not diverge exponentially from it. If a family is unstable, then neighboring trajectories will diverge exponentially from it and will wander over a region of phase space in general. If such an unstable orbit lies close to the asteroid, then the usual case is for the divergent trajectory to intersect the asteroid, or to suffer close approaches which send the trajectory away on a hyperbolic orbit. Figure 10 is a plot of the normalized Jacobi constant of a periodic orbit family member vs the periodic orbit periap-

sis, mean semi-major axis, and apoapsis. If the orbit is stable, it is represented by a solid line, and if unstable, by a dotted line. The periapsis, semi-major axis, and apoapsis of each orbit is drawn, so a horizontal line drawn from a particular value of the normalized Jacobi constant will intersect the periodic orbit periapsis first, the semi-major axis second, and the apoapsis third. Note that the retrograde orbits (negative Jacobi constant) are nearly circular and hence the three quantities nearly equal.

4.4. Ejection and Capture Dynamics

One of the most interesting aspects of dynamics about asteroids is the possibility for a given orbit to be ejected into a hyperbolic escape trajectory, or conversely for a hyperbolic orbit to be (temporarily) captured into an elliptic orbit. The possibility of these occurrences was first suggested to the authors by J. K. Miller (personal communication 1993). Such effects provide mechanisms for an asteroid to shed ejecta, capture nonejecta particles on its surface, and temporarily capture neighbors of the asteroid. These effects are driven primarily by the interaction between the particle orbit, the gravity C_{22} coefficient and the asteroid rotation rate.

4.4.1. Energy effects. The expected change in Keplerian energy and radius of periapsis and apoapsis due to particle interactions with the second order gravitational harmonics can be derived as follows. Restricting ourselves to the gravity potential of 2nd order only, which provides the major perturbations to the orbits,

$$U_2 = \frac{\mu}{r^3} \left[\frac{r_0^2 C_{20}}{2} (3 \sin^2 \alpha - 1) + 3r_0^2 C_{22} (1 - \sin^2 \alpha) \cos(2\lambda) \right], \quad (39)$$

where $\sin \alpha = \sin i \sin u$, $u = \nu + f$, and λ is the particle longitude in the body-fixed coordinate system.

Let us estimate the total variation one may expect in the apse radii from one periapsis passage. We assume that the orbital motion takes place in the equatorial plane, so $\sin \alpha = 0$, $\cos \alpha = 1$, and $\lambda = f + \nu - \omega t$, and assume that the equation is evaluated in the vicinity of periapsis. Later we shall perform a quadrature of the equation about the periapsis passage, so let us neglect any terms of the differential equations which are odd about the periapsis passage. Restricting ourselves thusly yields the following simplified formulas from Eqs. (29), (34), and (35),

$$\begin{aligned} \dot{C}_2 \approx & \frac{9\mu e r_0^2 C_{22} \dot{f}}{a(1-e^2)r^2} \sin 2\nu \sin f \sin 2(f - \omega t) \\ & - \frac{6\mu r_0^2 C_{22} \dot{f}}{r^3} \sin 2\nu \cos 2(f - \omega t) \end{aligned} \quad (40)$$

$$\begin{aligned} \dot{r}_p \approx & -\frac{9r_p^2 r_o^2 C_{22} \dot{f}}{a(1-e^2)r^2} \sin 2\nu \sin f \sin 2(f - \omega t) \\ & + \frac{6r_p^2 r_o^2 C_{22} \dot{f}}{er^3} \left(1 - \frac{r^2}{r_p^2}\right) \sin 2\nu \cos 2(f - \omega t) \end{aligned} \quad (41)$$

$$\begin{aligned} \dot{r}_a \approx & \frac{9r_a^2 r_o^2 C_{22} \dot{f}}{a(1-e^2)r^2} \sin 2\nu \sin f \sin 2(f - \omega t) \\ & - \frac{6r_a^2 r_o^2 C_{22} \dot{f}}{er^3} \left(1 - \frac{r^2}{r_a^2}\right) \sin 2\nu \cos 2(f - \omega t), \end{aligned} \quad (42)$$

where the true anomaly rate is $\dot{f} = \pm \sqrt{\mu a(1-e^2)}/r^2$, positive for direct motion and negative for retrograde motion.

To estimate the effect of a periapsis passage on these elements, let us introduce some assumptions which allow for a simple quadrature. First, assume that $r \approx r_p$ through the flyby, that $f \approx \dot{f}_p t$ around the flyby, and finally that the osculating elements are nominally conserved during the closest approach. This allows a quadrature about the periapsis from time $-T$ to time $+T$. Note the following results:

$$\int_{-T}^T \cos 2(\omega - \dot{f}_p)t \, dt = \frac{\sin 2(\omega - \dot{f}_p)T}{\omega - \dot{f}_p} \quad (43)$$

$$\begin{aligned} \int_{-T}^T \sin \dot{f}_p t \sin 2(\omega - \dot{f}_p)t \, dt = & -\frac{\sin(2\omega - 3\dot{f}_p)T}{2\omega - 3\dot{f}_p} \\ & + \frac{\sin(2\omega - \dot{f}_p)T}{2\omega - \dot{f}_p}. \end{aligned} \quad (44)$$

Performing the quadrature yields the formulas

$$\begin{aligned} \Delta C_2 \approx & -\frac{3\mu r_o^2 C_{22} \dot{f}_p}{r_p^3} \sin 2\nu \\ & \left[\frac{2 \sin 2(\omega - \dot{f}_p)T}{\omega - \dot{f}_p} + \frac{3e}{1+e} \left\{ \frac{\sin(2\omega - 3\dot{f}_p)T}{2\omega - 3\dot{f}_p} \right. \right. \\ & \left. \left. - \frac{\sin(2\omega - \dot{f}_p)T}{2\omega - \dot{f}_p} \right\} \right] \end{aligned} \quad (45)$$

$$\begin{aligned} \Delta r_a \approx & -\frac{9r_o^2 C_{22} \dot{f}_p}{a(1-e^2)} \left(\frac{r_a}{r_p}\right)^2 \sin 2\nu \\ & \left[\frac{4}{3} \left(1 + \frac{r_p}{r_a}\right) \frac{\sin 2(\omega - \dot{f}_p)T}{\omega - \dot{f}_p} + \frac{\sin(2\omega - 3\dot{f}_p)T}{2\omega - 3\dot{f}_p} \right. \\ & \left. - \frac{\sin(2\omega - \dot{f}_p)T}{2\omega - \dot{f}_p} \right] \end{aligned} \quad (46)$$

$$\begin{aligned} \Delta r_p \approx & \frac{9r_o^2 C_{22} \dot{f}_p}{a(1-e^2)} \sin 2\nu \left[\frac{\sin(2\omega - 3\dot{f}_p)T}{2\omega - 3\dot{f}_p} \right. \\ & \left. - \frac{\sin(2\omega - \dot{f}_p)T}{2\omega - \dot{f}_p} \right] \end{aligned} \quad (47)$$

$$\dot{f}_p = \pm \sqrt{\frac{\mu(1+e)}{r_p^3}}, \quad (48)$$

where in the current approximation $2T$ represents the time over which the particle will receive its largest transient perturbation. A value of $T = \pi/(4\omega)$ serves this purpose and corresponds to a time period of one-fourth of the asteroid revolution (when taken from $-T$ to $+T$). Note that these expressions give an estimate on the amplitude of the transient variations and not necessarily the total variation of these quantities.

To derive similar equations for an apoapsis passage, interchange all occurrences of $(-)_p$ and $(-)_a$, as well as changing the signs in front of the Δr_a and Δr_p equations. The magnitude of the effect at apoapsis is definitely smaller, and may often be neglected unless the orbit is near circular. For a retrograde orbit the angular rates ω and $\dot{f}_p$ add, so again the total magnitude of the variation is small.

4.4.2. General results on orbit dynamics. The above equations provide insight into the dynamics of a particle in a direct orbit about an asteroid. The first and most definite result is the relation between which quadrant (Fig. 11) of the asteroid contains the periapsis and changes in orbit energy. Every time a particle passes through periapsis, its argument of periapsis, denoted by ν , will lie in one of these quadrants. Now, note that the equation for ΔC_2 is proportional to the term $-\sin 2\nu$ (with the remaining terms being positive for most cases of interest). Thus, when the periapsis lies in quadrants I or III (i.e., $\nu \in [0, \pi/2]$, $[\pi, 3\pi/2]$), $\Delta C_2 < 0$ and the energy decreases. Conversely, when the periapsis lies in quadrants II or IV (i.e., $\nu \in [\pi/2, \pi]$, $[3\pi/2, 2\pi]$), $\Delta C_2 > 0$ and the energy increases. Significant increases and decreases in energy can occur and allow the asteroid to either capture a hyperbolic orbit that passes close enough to the body, or eject an elliptic orbit into a hyperbolic orbit that escapes from the asteroid. Thus a hyperbolic flyby with periapsis in quadrants I or III may be subject to capture, and an elliptic orbit with periapsis in quadrants II or IV may be subject to ejection.

For hyperbolic orbits, there is in general only one pass through the system and a single chance at capture. For elliptic orbits, if not ejected from the asteroid the orbits' argument of periapsis will be stochastically distributed about the asteroid from orbit to orbit. Even if some resonance originally exists in the orbit, the act of passing through periapsis will in general destroy the resonance that existed unless the argument of periapsis lies along the

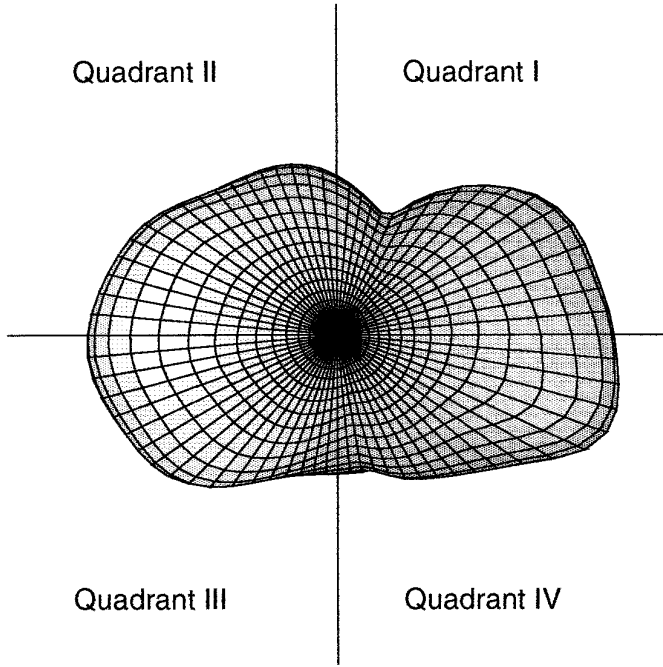


FIG. 11. Quadrants I–IV as defined for Castalia. The quadrant in which the orbit periapsis lies in (during periapsis passage) controls the type of perturbation the orbit receives. A passage in quadrants I or III decreases the Keplerian energy, while a passage in quadrants II or IV increases the Keplerian energy.

divisions between quadrants, where we find closed, periodic orbits close to the asteroid. In the general case the resonances cannot be preserved, due to the timing changes from orbit to orbit. Thus, in general, an elliptic orbit will encounter many opportunities to be ejected from the system. If the periapsis is in quadrants I or III, then the orbits' apoapsis is drawn in toward the asteroid, which may heighten the effect of the subsequent periapsis passage. Eventually, the periapsis will lie in quadrants II or IV, and ejection becomes a possibility. Note that a mitigating feature is that changes in the periapsis radius move in the opposite direction of the changes in the apoapsis radius. Thus, for a quadrant I or III passage, the orbit becomes more circular and, if the periapsis radius changes by a significant amount, the subsequent orbit's periapsis may be large enough to avoid some of the large changes. Conversely, for a quadrant II or IV passage the orbits' periapsis will decrease in general and subject the orbit to larger perturbations on ensuing passages.

4.4.3. Capture and escape radius. When can one periapsis passage yield a large enough change in the Keplerian energy to either capture a hyperbolic orbit or eject an elliptic orbit? A parabolic orbit has zero Keplerian energy, so substituting the proper terms in

$$\Delta C_2 + C_2 = 0 \quad (49)$$

yields the condition

$$0 = 1 - e^2 + \frac{6r_0^2 C_{22} \dot{f}_p \sin 2\nu}{r_p^2} \left[\frac{2(1+e) \sin 2(\omega - \dot{f}_p)T}{\omega - \dot{f}_p} + 3e \left\{ \frac{\sin(2\omega - 3\dot{f}_p)T}{2\omega - 3\dot{f}_p} - \frac{\sin(2\omega - \dot{f}_p)T}{2\omega - \dot{f}_p} \right\} \right], \quad (50)$$

where

$$\dot{f}_p = \sqrt{\frac{\mu(1+e)}{r_p^3}}. \quad (51)$$

It is immediately clear that for a hyperbolic orbit to be captured its periapsis must lie in quadrants I or III. Conversely, for an elliptic orbit to be ejected its periapsis must lie in quadrants II or IV. The above equation defines one curve in the (r_p, e) plane for each value of ν . Orbits which lie between this curve and $e = 1$ may be subject to either capture or ejection during any periapsis passage in the proper quadrant. It is important to note that an orbit lying in these regions is not guaranteed to be captured or ejected. Also, an elliptic orbit outside of the ejection region may still be ejected after several periapsis passages. Figure 12 presents this curve, the lower curve being defined for

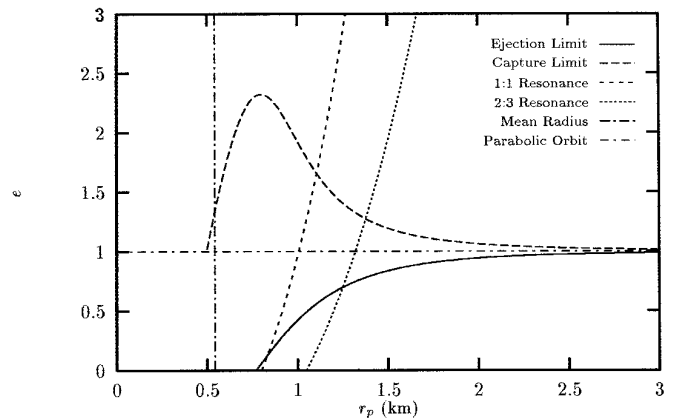


FIG. 12. Conditions for particle capture and ejection at Castalia. Passing hyperbolic orbits that lie in the envelope and have periapsis in quadrants I or III may be captured into orbit around Castalia; the strongest capture is expected close to the resonance lines. Elliptic orbits that lie in the envelope and have periapsis in quadrants II or IV may be ejected in one periapsis passage, and again the strongest ejection is expected close to the resonance lines. Elliptic orbits not in the envelope may still be ejected in future orbits.

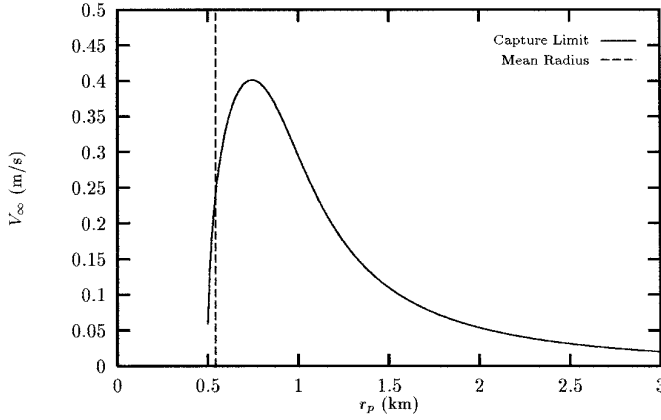


FIG. 13. Limiting V_∞ for particle capture at Castalia. For Castalia only extremely low velocity intruders are susceptible to capture.

passage through quadrants II or IV, and the upper curve for passage through quadrants I or III.

Figure 13 provides additional details for the capture conditions and plots the capture curve condition in terms of the periastron radius and V_∞ of the particle. For Castalia the maximum capture V_∞ is less than ~ 0.4 m/sec (which is very small). For larger asteroids this number may grow appreciably.

4.4.4. Resonance effects. The 1:1 and 2:3 resonance lines in Fig. 12 correspond to the periastron radius–eccentricity combinations which yield these resonance ratios between the asteroid rotation rate and the particle true anomaly rate at periastron. Note that these resonances are only satisfied over the short time span during the particle’s periastron passage, yet one such passage at a resonance ratio may yield significant changes to the orbit energy and eccentricity. In the current formulation, the resonant term becomes proportional to the quadrature time T , thus the results are somewhat conservative. It is expected that the change in orbital energy will be greatest when the particle lies along one of these resonance lines. Thus it is instructive to discuss them for a moment.

Consider a hyperbolic orbit which has a Castalia flyby within the capture line and along the 2:3 resonance line. Assuming that the particle is captured, its eccentricity will drop to less than 1 and its argument of periastron will increase (both amounts can be estimated). Thus, after capture around the 2:3 resonance line, the orbit will move away from the resonance line, perhaps even into the region where direct ejection is not possible. In subsequent periastron passages the apoapsis and periastron changes are not expected to be as large, since the particle is not near the resonance line. Hence, such a capture may lead to a long-term capture of the particle in question, although the possibility for eventual escape can never be discounted.

Conversely, if a particle becomes captured at the 1:1

resonance, the subsequent orbit will lie between the two resonance lines (in the eccentricity–periastron radius space). This particle will be much more likely to suffer ejection, as it lies between two regions where the characteristic changes in orbit energy and eccentricity will be large in general.

4.5. Retrograde Orbits

Retrograde orbits can be analyzed in terms of classical planetary orbiter results. In this situation the semi-major axis, eccentricity, and inclination have only short-period variations. The argument of periastron, argument of the ascending node, and mean epoch all have a secular variation which is influenced most strongly by the gravity coefficient C_{20} . See Scheeres (1994) for a more in-depth discussion of this case, including higher order effects of the gravity field.

For Castalia the evolution of the mean osculating elements is approximated by

$$a = a_o \quad (52)$$

$$i = i_o \quad (53)$$

$$e = e_o \quad (54)$$

$$\Omega = \Omega_o - 3.35 \times 10^{-5} \frac{t \cos i}{a^{7/2}(1 - e^2)^2} \quad (55)$$

$$\omega = \omega_o - 3.35 \times 10^{-5} \frac{t}{a^{7/2}(1 - e^2)^2} \left[\frac{5}{2} \sin^2 i - 2 \right] \quad (56)$$

$$M_o = M_{o_o} + t \left(\frac{3.07 \times 10^{-4}}{a^{3/2}} - 3.35 \times 10^{-5} \frac{t}{a^{7/2}(1 - e^2)^{3/2}} \left[\frac{3}{2} \sin^2 i - 1 \right] \right) \quad (57)$$

where the time is measured in seconds and the semi-major axis in km. For example, a quasi-circular orbit with radius 1 km will have a secular nodal rate up to $\sim 166^\circ/\text{day}$, which is quite large.

Note that if the periastron radius of the orbit becomes small and the eccentricity becomes large, then the retrograde orbits may also be subject to some of the large perturbations felt by the direct orbits. In these situations one also observes the onset of chaotic dynamics. Note that, from the retrograde family of periodic orbits, near circular orbits are definitely stable up to a radius of 1 km and are possibly long-term stable within a radius of ~ 0.9 km.

5. DYNAMICS OF SURFACE PARTICLES

In this section we derive the local accelerations acting on a particle as a function of surface location on Castalia by computing the local slope at each point of its surface.

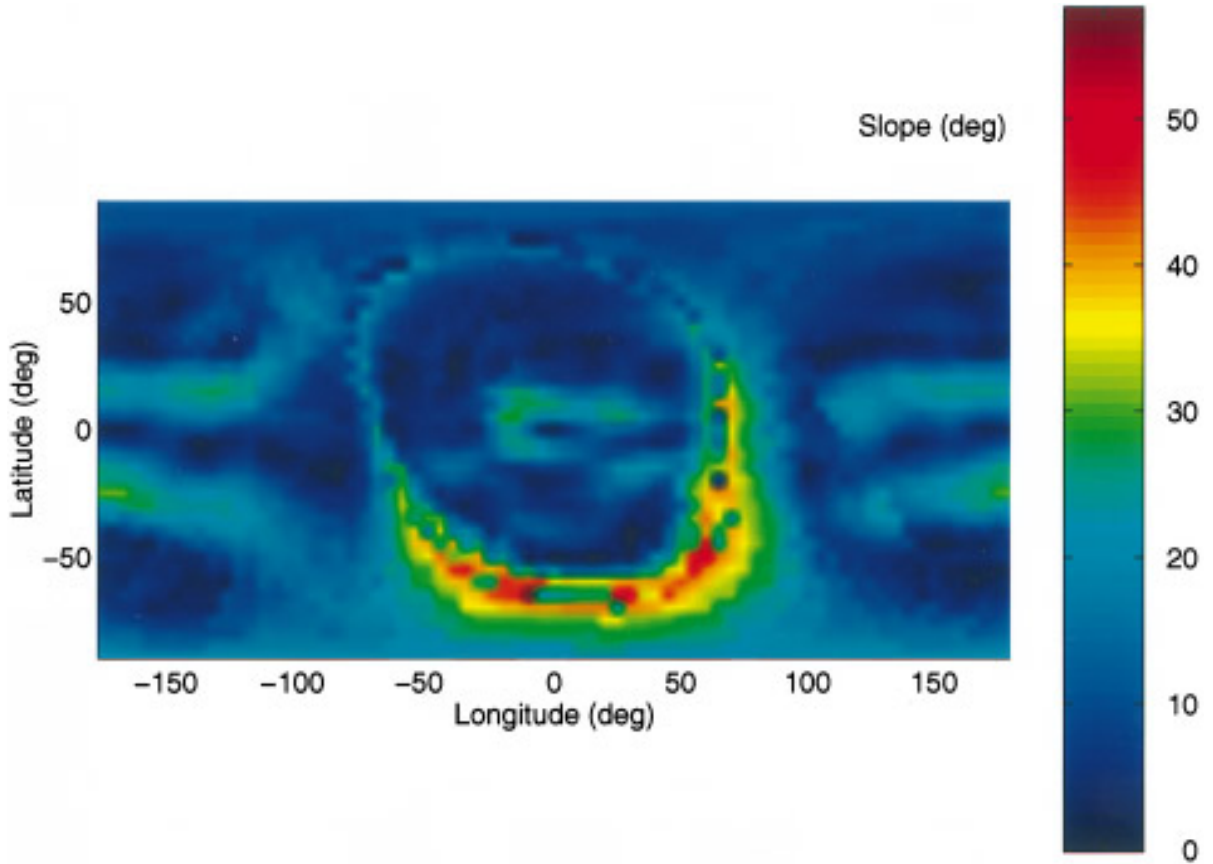


FIG. 14. Slopes over the surface of Castalia. The steepest gradients go into the crevice between the two lobes.

Then we investigate the dynamics of ejecta launched from the surface of the asteroid. It is desirable to characterize the final evolution of an ejected particle given its surface location and launch velocity, and to characterize where on the surface ejecta may accumulate. This paper addresses the first concern in detail and discusses the second concern in light of the results found herein (see Dobrovolskis and Burns 1980 and Geissler *et al.* 1994 for a further discussion of the second effect). The Jacobi integral is used to derive a discriminant quantity which identifies whether or not a particle orbit may have come from the surface of the asteroid in the past, or if it may impact on the surface of the asteroid in the future. Next, some brief comments on the possible modes in which a particle may become trapped

into stable orbits about the asteroid are made. Finally we discuss the dynamics and distribution of re-impact ejecta.

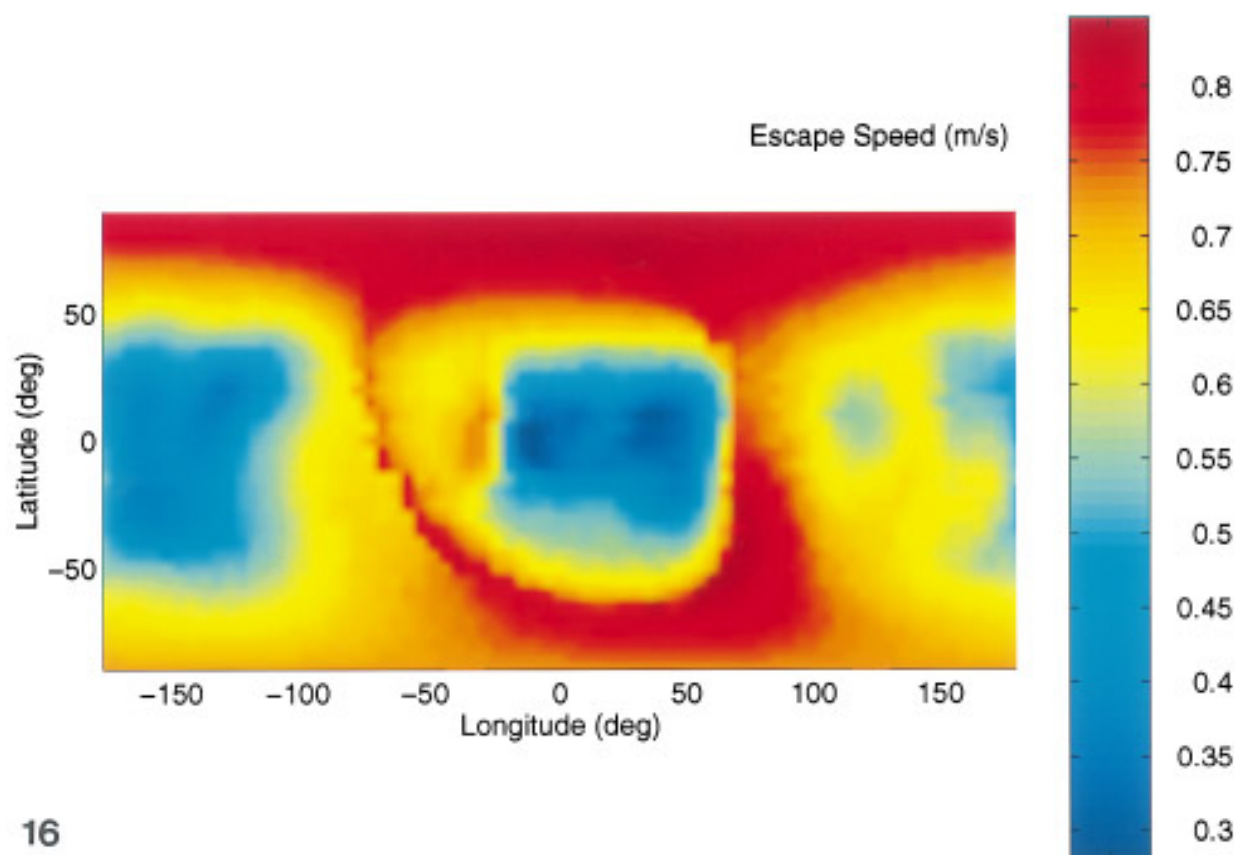
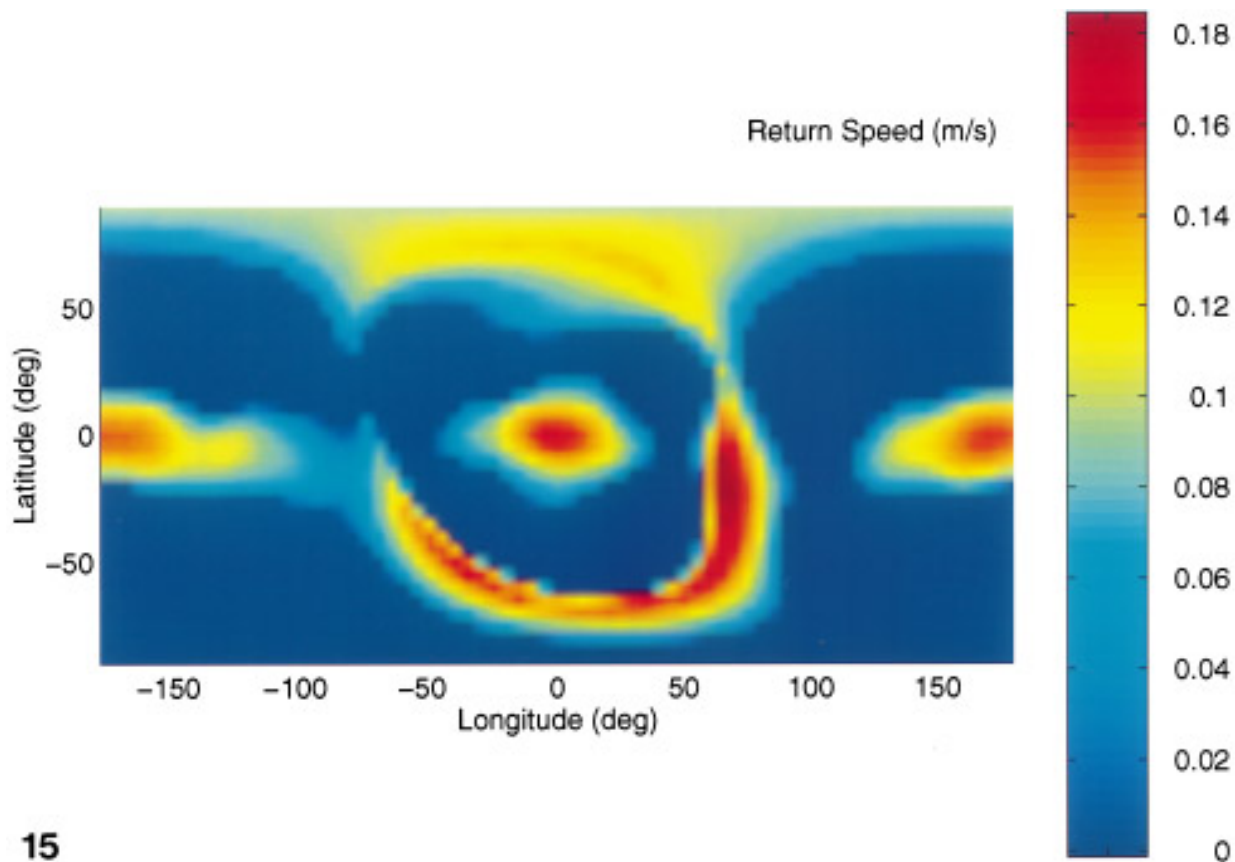
5.1. Surface Forces

The force acting on a particle at each location on the surface of the asteroid arises from both the gravitational and centripetal accelerations and is computed from Eq. (19) to be the vector V_r . The dynamic slope on the surface of the asteroid (ϕ) is related to the angle between the normal vector ($\mathbf{n}$) and the force vector at a given surface location.

$$\cos(\pi - \phi) = \frac{\mathbf{n} \cdot V_r}{|V_r|} \quad (58)$$

FIG. 15. Guaranteed return speeds over Castalia's surface. If the launch speed is less than the given speed, the particle will return to the surface (note regions of zero speed; in these areas the criterion is too conservative to yield bounds. These regions would disappear if a larger asteroid density were used).

FIG. 16. Local normal escape speed over Castalia's surface. If the launch speed is greater than the given speed the particle will escape. These numbers are derived assuming the particle is launched normal to the surface.



If the slope $\phi = 0^\circ$, then the force and normal vectors are anti-parallel. If the slope $\phi > 90^\circ$, then a loose particle on the surface will be thrown off of the asteroid. For intermediate slopes the motion of a loose particle is controlled by the coefficient of friction between the particle and the asteroid surface.

If a coefficient of bulk friction is assumed for a particular location on the surface (μ_F), it is possible to compute the smallest angle (the limiting slope) for which a particle will be subject to sliding (Greenwood 1988):

$$\mu_F = \tan \phi. \quad (59)$$

Thus, for an infinitely rough surface ($\mu_F = \infty$), the limiting slope is 90° , while for a perfectly smooth surface ($\mu_F = 0$), the limiting slope is 0° . Whereas this bulk characterization of friction is grossly simplified and does not account for the actual physics of sliding or rolling friction, it does provide a simple way to assign a bulk number to the surface in question.

For the assumed density of 2.1 g/cm^3 the maximum slope over the surface of Castalia is 57° (which occurs at latitude of -64° and longitude of -91° near the crevice between Castalia's lobes). Figure 14 shows the contours of the slope over the surface of Castalia for a density of 2.1 g/cm^3 . Given this maximum slope, a friction coefficient of at least ~ 1.6 would be required for a nonsliding condition to hold over the entire surface of Castalia. Previous studies have assumed a maximum slope of 33° before slipping occurs (Thomas *et al.* 1994), corresponding to a friction coefficient of 0.65. Thus, for the current Castalia model, there are implied slopes large enough to allow for the migration of regolith. An interesting question is what the minimum density of Castalia would be before all surface slopes are less than 33° . Our computation, taken over the entire surface, indicates a necessary density much larger than feasible or expected (by an order of magnitude), indicating that Castalia definitely has slopes greater than 33° .

Another quantity of interest is the minimum Castalia density for which particles are not thrown off the ends of the asteroid. This density may be computed at each point of the surface by finding the necessary density for the local normal and the local force vectors to be orthogonal. For the Castalia shape and rotation model this minimum density is 1.3 g/cm^3 and occurs at the longest end of the asteroid, as expected. This places a lower bound on the density of an unconsolidated asteroid (Burns 1975).

5.2. Guaranteed Return Speed

It is possible, by using the existence of the zero-velocity curves, to derive an upper value on ejecta speed which ensures that all ejecta with speeds less than this value will eventually fall back onto the surface. This upper bound is

conservative. Consider the zero-velocity curves and the $\pm S$ equilibrium points discussed previously. For values of C greater than $C_{\pm S}$ the zero-velocity curves divide the space around the asteroid into at least two distinct regions, one connected to infinity and the other(s) containing the asteroid. For the Castalia shape with a density of 2.1 g/cm^3 , the zero-velocity curves still intersect the Castalia shape when $C \approx C_{\pm S}$, thus there are two disjoint components of this zero-velocity surface which contain the two ends of the asteroid. Any ejecta launched from the surface of Castalia with a Jacobi constant $C > C_{\pm S}$ will stay trapped close to the surface and will eventually re-impact the asteroid. Using this condition we can derive, as a function of position on the Castalia surface, the maximum ejecta speed which will guarantee that the ejecta will fall back onto the asteroid. For speeds greater than this speed it becomes necessary to investigate the individual trajectories to see whether or not they fall back to the surface. This speed is derived by solving the equation $J(\mathbf{r}, \dot{\mathbf{r}}) = -C_{\pm S}$ for the asteroid relative ejecta speed $v = |\dot{\mathbf{r}}|$:

$$v = \sqrt{2(V(\mathbf{r}) - C_{\pm S})}. \quad (60)$$

Figure 15 presents a contour plot of the local guaranteed return speed. The limiting speeds are larger at the asteroid ends and in the crevice.

5.3. Local (Normal) Escape Speed

The necessary escape speed over the surface of Castalia is another item of interest as it defines threshold values below which the ejecta may re-impact the asteroid surface. A precise computation of these thresholds is nontrivial as the boundary between escape and re-impact (as a function of initial surface speed) is expected to be complicated. To simplify the characterization we compute the necessary speed normal to the local surface slope for the particle to have reached escape velocity in inertial space. To account for local variations in the gravity field that yield a larger effective potential, the escape condition is presumed to be met when the inertial velocity of a particle leaving the surface is greater than $\max[\sqrt{2U(\mathbf{r})}, \sqrt{2\mu/|\mathbf{r}|}]$. In numerical checks on this condition, it appears to be a reliable sufficiency condition for particle escape to occur.

Assume that a particle leaves the asteroid normal to the local surface, or $\mathbf{v} = v\mathbf{n}$ where $\mathbf{n}$ is the local normal to the surface and v is the ejecta speed. The inertial velocity of such a particle is

$$\mathbf{v}_I = v\mathbf{n} + \Omega \times \mathbf{r}, \quad (61)$$

where $\mathbf{r}$ is the vector to the surface and Ω is the asteroid's angular velocity vector. To find the local escape speed, set the magnitude of the inertial velocity equal to $\sqrt{2U_{\max}} =$

$\max[\sqrt{2U(\mathbf{r})}, \sqrt{2\mu/|\mathbf{r}|}]$. Evaluating this condition and solving the resultant quadratic equation yields

$$v = -\mathbf{n} \cdot (\Omega \times \mathbf{r}) + \sqrt{[\mathbf{n} \cdot (\Omega \times \mathbf{r})]^2 + 2U_{\max} - (\Omega \times \mathbf{r})^2}. \quad (62)$$

This quantity has been computed over the surface of Castalia using the polyhedral gravitational field (Fig. 16). Over most of the surface the escape speed lies between 0.55 and 0.84 m/sec, at the long ends of the asteroid (around longitude 0° and 180°) the speeds lie between 0.28 and 0.55, and in the crevice between Castalia's two lobes and at its north pole the speeds are over 0.75 m/sec.

5.4. Constraints on the Origin of an Orbiting Particle

A criterion may be developed which can discriminate whether or not a particle's trajectory could either have emanated from the asteroid surface or will intersect the asteroid surface in the future. This criterion relies on the Jacobi integral and is applicable to direct, low inclination orbits.

To establish the criterion, the Jacobi integral must first be expressed in terms of osculating orbital elements. Assume that this evaluation occurs when the particle orbit crosses the equatorial plane of the asteroid, as defined by the plane normal to the rotation pole and containing the asteroid barycenter. Then, expressing the particle orbit in terms of osculating elements we find

$$C = \frac{\mu}{2a} + \omega\sqrt{\mu a(1-e^2)} \cos i + \sum_i U_i(\mathbf{r}), \quad (63)$$

where U_i contains all higher order gravitational terms. In the following analysis, it is found that if these higher order terms are neglected, the results are changed on the order of 5%. Thus, for generality in the following only the zeroth order gravitational potential is accounted for. If desired for a particular case, a more accurate analysis could be performed. Neglecting the higher order terms yields the approximate relationship

$$C \approx \frac{\mu}{2a} + \omega\sqrt{\mu a(1-e^2)} \cos i. \quad (64)$$

Next consider the Jacobi integral itself, $C = V - T_E$. Evaluating the Jacobi constant over the entire asteroid surface (assuming $T_E = 0$) yields a maximum value C^* of the Jacobi constant C . For Castalia $C^* = 2.1223 \times 10^{-7}$ km²/sec² and occurs at a latitude of -17° and longitude of 67° . If a particle trajectory intersects the asteroid surface, that trajectory must have a Jacobi constant less than C^* , as can be easily inferred from the relation $C = V - T_E$

and $T_E \geq 0$. Thus if a particle orbit has a Jacobi constant C greater than C^* , it does not intersect with the asteroid surface in either the future or the past.

Using the relation Eq. (64), we can explicitly relate a given orbit about the asteroid with the criterion for whether it emanated from the asteroid surface. If the inequality

$$\frac{\mu}{2a} + \omega\sqrt{\mu a(1-e^2)} \cos i > C^* \quad (65)$$

holds, then the particle's trajectory could not have originated from the asteroid surface. Usually the relationship $C^* > \mu/(2a)$ holds. Thus whenever the inclination exceeds 90° the inequality is trivially false and the criterion no longer applies. In fact, as the particle inclination progressively grows from 0° to 90° , the applicability of this criterion decreases. Thus, in practice, this criterion is only useful for nearly equatorial, direct orbits. Note that Dactyl is in such an orbit about Ida (Belton *et al.* 1995). Allowing for the small angle approximation, it would be valid to set $\cos i \approx 1$ for $i < 10^\circ$.

Now define two characteristic lengths,

$$\alpha = \frac{1}{2\mu} \left(\frac{C^*}{\omega \cos i} \right)^2 \quad (66)$$

$$\beta = \frac{\mu}{C^*}, \quad (67)$$

where both of these values have units of kilometers. For Castalia, these constants are found to be

$$\alpha = 1.303/\cos^2 i \text{ km} \quad (68)$$

$$\beta = 0.443 \text{ km}. \quad (69)$$

The inequality can be expressed in several different ways; the two of immediate interest are

$$(1-e^2)a^3 - 2\alpha a^2 + 2\alpha\beta a - \frac{1}{2}\alpha\beta^2 > 0 \quad (70)$$

$$(r_a - \alpha)r_p^2 + (r_a^2 - 2\alpha r_a + 2\alpha\beta)r_p - \alpha(r_a - \beta)^2 > 0. \quad (71)$$

A few quick notes on these inequalities: Condition (70) is a cubic and is useful for evaluating the condition when $e = 0$. Condition (71) is also valid if r_a and r_p are interchanged, and can be readily factored using the quadratic equation

$$r_p > \frac{r_a^2 - 2\alpha r_a + 2\alpha\beta}{2(r_a - \alpha)} \left[\sqrt{1 + \frac{4\alpha(r_a - \beta)^2(r_a - \alpha)}{(r_a^2 - 2\alpha r_a + 2\alpha\beta)^2}} - 1 \right]. \quad (72)$$

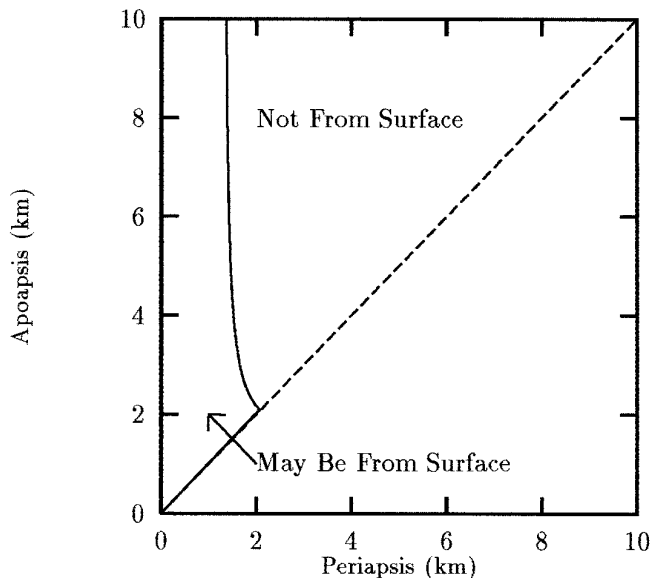


FIG. 17. Ejecta exclusion criterion for $i = 0$. Given an orbit, this plot allows for an immediate classification to be made, whether or not the trajectory may ever intersect the asteroid (in the future or past).

This inequality is plotted in Fig. 17 for $i = 0$ from $r_a = r_p$ to $r_a \rightarrow \infty$. Taking the limit $r_a \rightarrow \infty$ yields the limiting inequality $r_p > \alpha$. In the figure, any particle whose orbit which falls to the right of the solid line could not have come from the asteroid surface and any particle whose orbit falls to the left of the solid line may have come from the asteroid surface.

An important point to note in the plot is that all quasi-circular orbits outside of ~ 2 km could not have originated from Castalia's surface. Recall that orbits with a mean semi-major axis within 1.4 km of Castalia's surface are unstable. Thus, this leaves only a relatively small range of direct orbits which are stable and which potentially come from the surface of the asteroid.

5.5. Ejecta Transport into Orbit

Consider the possibility for ejecta to depart the surface of the asteroid and end in a direct trajectory which is "stable" for some length of time. The direct family of periodic orbits provides a useful measure of how close to the asteroid surface stable orbits exist. For Castalia, members of this orbit family with periapsis inside ~ 1.3 km are unstable (mean semi-major axis within ~ 1.4 km). This places some limitations on the region where ejecta could become trapped in a stable orbit, depicted in Fig. 18.

The necessary condition for ejecta to be placed into this stable region is that the periapsis radius be raised to at least ~ 1.3 km. The mechanism for this to occur is during an apoapsis passage of the orbit through quadrants I or

III of the asteroid. The amount of "kick" in the periapsis radius due to one apoapsis passage may range up to order 0.5 km assuming $r_a = 1$ km and $r_p = 0.5$ km. However, after such a large change in periapsis subsequent changes will be smaller due to the higher periapsis and lower eccentricity. Thus, even after such a large change in periapsis a number of successive additive changes would still be required to raise the periapsis into the stable region. Note that these same orbits will be subject to changes in apoapsis radius on the order of kilometers during each periapsis passage, thus implying that it is much more likely that an ejection of the particle from the Castalia system would occur.

For retrograde orbits the situation is different. Note that the family of retrograde orbits extends all the way down to the asteroid surface, with stable orbits existing within hundreds of meters from the asteroid surface. In this situation the ejecta orbits are not susceptible to large changes in periapsis and apoapsis radius, and thus a mechanism like that for direct orbits may not apply.

A more likely scenario follows. Consider that a meteoritic impact may effectively disperse a portion of the asteroid into some velocity distribution. All that is required is that some members of the velocity distribution be given a speed of ~ 0.7 m/sec in a retrograde direction, and that the initial radius of the orbit, as measured from the asteroid center of mass, is large enough to avoid the longer ends of the resulting asteroid. Such an orbit is stable and may exist for long periods of time. This scenario implicitly implies that one of the longer ends of the asteroid is removed in the collision.

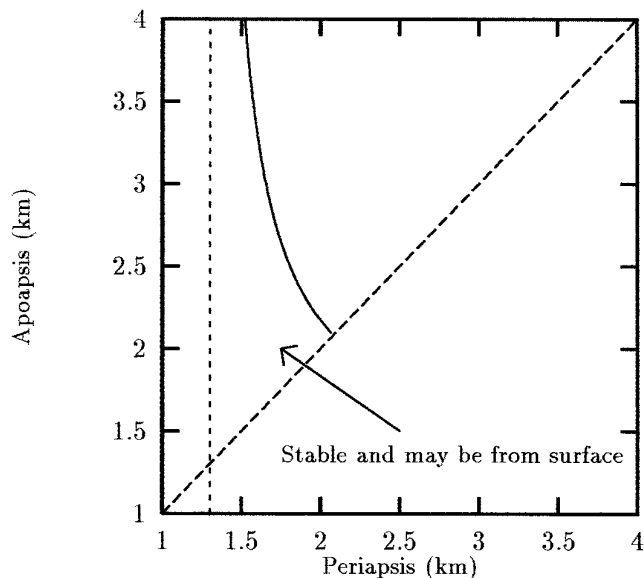


FIG. 18. Stable region of direct orbits that may originate from the surface.

5.6. Distribution of Return Ejecta

An important issue to address is the expected distribution of return ejecta over the surface of Castalia. There are two approaches one may take to derive estimates of these distributions. The first, and simplest, is to identify regions of the Castalia surface which have large slopes (as defined earlier). Return ejecta in these regions will be susceptible to migration. The second is to characterize the trajectories of ejecta and map these back to the surface again.

5.6.1. Surface distribution. Note from Fig. 14 that the regions of lowest slope range over the north pole and cover the regions of each lobe half-way between the long ends and the crevice. The southern pole has a uniformly higher slope. In the vicinity of the long ends of the asteroid (around longitude 0° and 180°), the slopes tend to be higher, although they do not exceed 30° , so one may not expect much migration away from the ends of the asteroid. Indeed, the ends of Castalia seem to be blunted, perhaps indicating that natural processes have already shaped the ends of the asteroid.

Over the southern hemisphere one expects a more dynamic situation with regolith flow into the crevice, which exhibits the largest slopes. In the interior of the crevice these slopes exceed 45° , large enough for one to expect the free migration of any loose regolith. In these regions, however, one would expect the regolith to flow into the crevice and, over time, reduce its slope. Thus, whatever the current slope structure of the crevice, it would grow steeper as one moves back in time to the original event when the two lobes came into contact. For the current Castalia the flow of regolith would go from the southern pole, from the longitude area around $\sim 70^\circ$, and potentially from the latitudes just south of the 180° longitude into the crevice which extends from the south of the 0° longitude in a semi-circle (in latitude-longitude space) to the equator at $\sim 70^\circ$. Note that the two lobes are widely separated there, so an influx of regolith may not be adequate to “fill-in” this crevice, even over long time spans.

5.6.2. Dynamics of ejecta. The distribution of return ejecta is a very complex problem for several reasons. First, the parameter space over which one must search to generate global results (latitude, longitude, and initial velocity vector) is very large. Second, given the severe distortion in Castalia’s gravity field analytical methods for mapping the ejecta dynamics close to the surface will not give valid results. The “brute force” approach to investigating this question would be to perform a Monte Carlo analysis where the parameter space is sampled with some bounds on the velocity sampling. Some work has been done with this approach by other authors (Geissler *et al.* 1995) using

a simple shape model (a tri-axial ellipsoid). We plan to perform a similar analysis with the current model in the future, probably using a parallel computer. Geissler *et al.* noted the tendency of the re-impact ejecta to accumulate on the “leading-edges” of the asteroid Ida. Moreover, the ejecta which tended to accumulate in these regions had initial speeds near escape speed.

For the Castalia model the leading edges would be longitudes in the approximate ranges $0^\circ \rightarrow 45^\circ$ and $180^\circ \rightarrow 225^\circ$. That a uniformly falling field of ejecta would tend to re-impact along the leading edges of the asteroid can be seen using a simple analysis under the assumption that the ejecta leave the surface near escape speed. As viewed from inertial space the ejecta orbits will then have larger eccentricities and hence apoapsis radii far from the asteroid. Thus, after leaving the asteroid surface, they will travel far enough from the asteroid so that their orbits will be qualitatively similar to standard Keplerian ellipses. On their initial return to the asteroid they may be crudely modeled as infalling particles on nearly straight lines, fixed in inertial space. Given such a situation, it is obvious that the leading edges of the rotating asteroid will, as a consequence of their travel, move into these infalling orbits as time progresses. The trailing edges of the asteroid will, instead, move away from these infalling orbits. Thus, one would expect to see a bias toward impacts on the leading edges of an asteroid, at least when considering ejecta leaving close to escape speeds.

When the ejecta speeds are lower the dynamics become much more complicated and such a simple analysis does not apply. Castalia escape speeds are very small (Fig. 16), much less than the expected impact velocity of an interloper which causes the initial ejecta field (Bottke *et al.* 1994). Thus only a small fraction of ejecta will potentially fall back onto the asteroid. The fraction of ejecta whose speeds are less than escape speed will follow an essentially chaotic orbit about the asteroid until they either escape or return. De Jong and Suzuki (1996) animate a number of specific trajectories about Castalia, including a return trajectory and an escaping trajectory. Both of these trajectories are very complex, and both will change substantially given slightly different starting conditions. Their complex nature highlights the difficulties in arriving at general results concerning ejecta re-distribution for low launch speeds.

6. CONCLUSIONS

All the results in this paper were possible due to the existence of the radar-derived shape model of Castalia. The majority of the computations carried out in this paper can easily be generated given the model of any arbitrary uniformly rotating asteroid. Indeed, we expect an increas-

TABLE I
Castalia Gravity Field Coefficients through Order 4

Order	Degree	C coefficient	S coefficient
0	0	1.0	—
1	0	0.0	—
1	1	0.0	0.0
2	0	-0.110298	—
2	1	0.0	0.0
2	2	0.156733	0.0
3	0	-0.015112	—
3	1	-0.037935	0.001211
3	2	0.006325	0.000616
3	3	0.020568	-0.013715
4	0	0.036630	—
4	1	0.002706	0.000407
4	2	-0.051363	0.003949
4	3	0.006140	-0.001747
4	4	0.050334	-0.006839

ing number of radar-derived shape models of asteroids to become available during the next few years.

Several areas, such as predictions of the asteroid's capture and ejection radii, need further investigation. Issues here include the probability of capture and the likely lifetime of a captured particle. Also, the potential role of the resonances must be addressed, as they may significantly influence this phenomenon.

The distribution of impact ejecta over the surface of an asteroid is a subject that also deserves a great deal of attention. Future research can focus on two main areas. The first would expand upon the analytical approaches derived in this paper to estimate the final evolution of ejecta. The other would approach the problem as a numerical experiment. This would entail Monte Carlo simulations taken over the entire asteroid surface and simulations of the evolution of specific impact velocity fields.

APPENDIX: GRAVITY COEFFICIENTS

Following are the Castalia gravity coefficients through order 4 (shown in Table I). These coefficients are normalized, as defined in (Kaula 1966), computed with respect to a normalizing radius $r_o = 0.5431$ km. The general form of the gravitational field can be expressed as

$$U = \sum_{l=0}^{\infty} \sum_{m=0}^{\infty} \left(\frac{r_o}{r} \right)^{l+1} P_{lm}(\sin \phi) [C_{lm} \cos m\lambda + S_{lm} \sin m\lambda], \quad (73)$$

where P_{lm} are the normalized Legendre polynomial, ϕ is the particle latitude, and λ is the particle longitude in the body-fixed frame.

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